

3rd Aug.
2020
Mon
Q-1

Assigned -2

lab

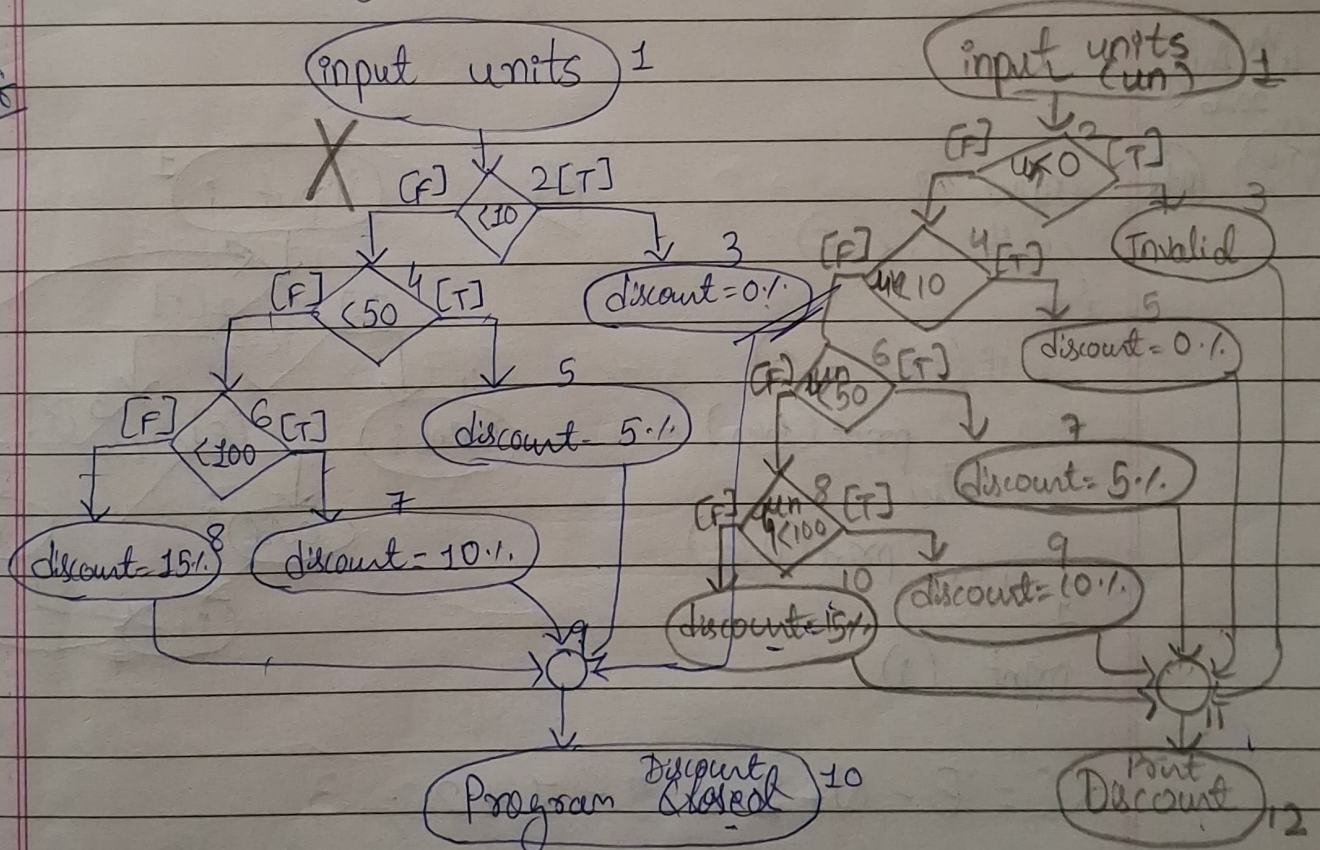
```

input units;
if units < 10
    print 'discount = 0%'
else if units < 50
    print 'discount = 5%'
else if units < 100
    print 'discount = 10%'
else
    print 'discount = 15%'
end if
print ('Discount Program Closed');
    
```

5
2

Unit-70
10-

Flow



CC's

$$CC = E - N + 2$$

$$= 15 - 10 + 2 = 3 + 2 = 5$$

$$CC = D + 1$$

$$= 4 + 1 = 5$$

10th Aug,
2026
Mon

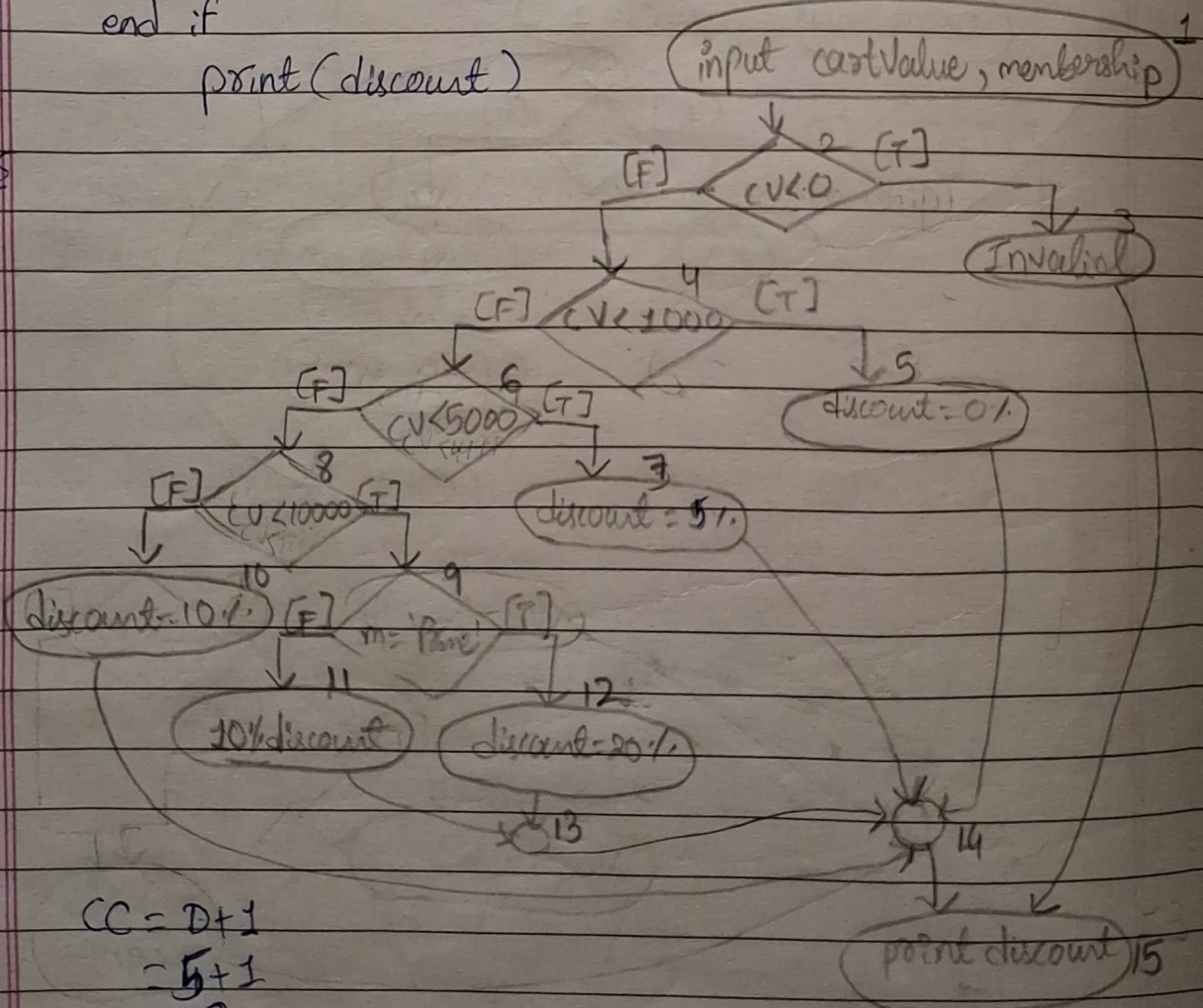
Assignment-2

2) ~~write (y)~~
input

```

input costValue, membership;
if costValue < 1000
    print(discount = 0%)
else if costValue > 1000 OR costValue < 4999
    print(discount = 5%)
else if costValue > 5000 OR costValue < 9999
    print(discount = 10%)
else if costValue > 10000 AND membership = 'Prime'
    print(discount = 20%)
else
    print('insert valid inputs')
end if
print(discount)
    
```

FGIS



$$\begin{aligned}
 CC &= D+1 \\
 &= 5+1 \\
 &= 6
 \end{aligned}$$

①

Path 1 = 1-2-3-11-12

Path 2 = 1-2-4-5-11-12

Path 3 = 1-2-4-6-7-11-12

Path 4 = 1-2-4-6-8-9-11-12

Path 5 = 1-2-4-6-8-10-11-12

① Test Case ID	Input	Expected message	Remarks
TC01	un = -5	Invalid input	Path 1 followed
TC02	un = 8	discount = 0%	Path 2 followed
TC03	un = 49	discount = 5% applied ∴ discount = 2.45	Path 3
TC04	un = 99	discount = 10% applied ∴ discount = 9.9	Path 4
TC05	un = 505	discount = 15% ∴ discount = 75.75	Path 5

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Assignment-2

while (1)
(2) { input

input cartValue, membership;

if cartValue < 1000

print(discount = 0%);

else if cartValue > 1000 ~~1~~ ^{or} cartValue < 4999

print(discount = 5%);

else if cartValue > 5000 ~~2~~ ^{or} cartValue < 9999

print(discount = 10%);

else if cartValue > 10000 ~~3~~ ^{AND} membership = 'Prime'

print(discount = 20%);

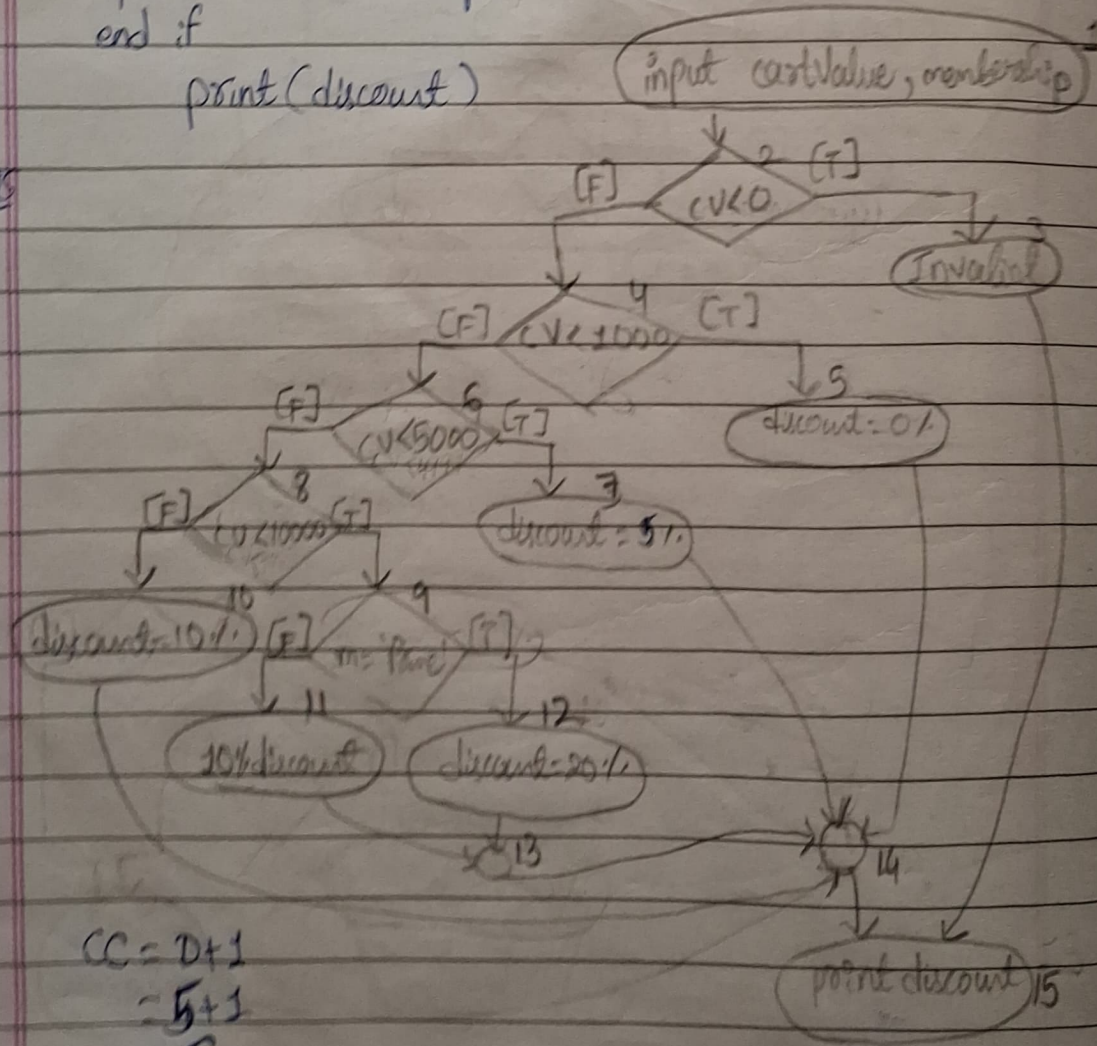
else

print('insert valid inputs')

end if

print(discount)

Flow



$$\begin{aligned}
 CC &= D + 1 \\
 &= 5 + 1 \\
 &= 6
 \end{aligned}$$

②

Path 1 = 1 - 2 - 3 - 14 - 15

Path 2 = 1 - 2 - 4 - 5 - 14 - 15

Path 3 = 1 - 2 - 4 - 6 - 7 - 14 - 15

Path 4 = 1 - 2 - 4 - 6 - 8 - 9 - 12 - 13 - 14 - 15

Path 5 = 1 - 2 - 4 - 6 - 8 - 9 - 11 - 13 - 14 - 15

Path 6 = 1 - 2 - 4 - 6 - 8 - 10 - 14 - 15

<u>Test Case ID</u>	<u>Input</u>	<u>Expected message</u>	<u>Remarks</u>
TC01	cu = -5 m = 'Prime'	Invalid input	Path 1
TC02	cu = 500 m = 'NonPrime'	discount = 0%	Path 2
TC03	cu = 4898 m = 'NonPrime'	discount = 5% = 244.9	Path 3

TC04

CV = 19500
m = 'NonPrime'

discount = 10%
= 1950

Path 5

TC05

CV = 9500
m = 'NonPrime'

discount = 10%
= 950

Path 6

TC06

CV = 19500
m = 'Prime'

discount = 15%
= 2925

Path 4

③

input units

if units ≤ 100

bill = units * 0.75

else if units ≤ 300

bill = 75 + (units - 100) * 1.00

else if units ≤ 500

bill = 275 + (units - 300) * 1.5

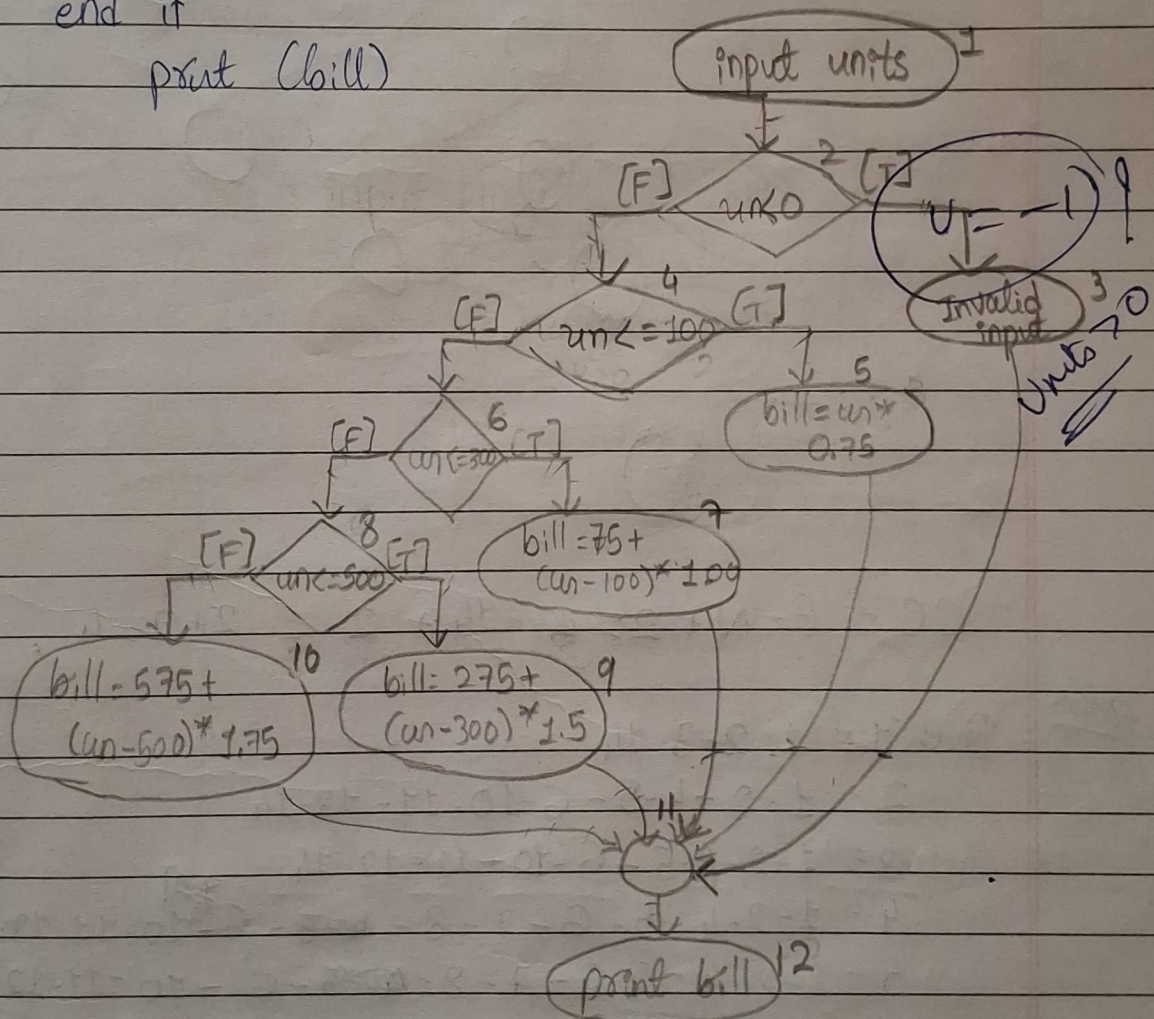
else

bill = 575 + (units - 500) * 1.75

end if

print (bill)

FGS



$$\begin{aligned}
 CC &= E - N + 1 &= D + 1 \\
 &= 15 - 10 + 1 &= 4 + 1 \\
 &= 6 &= 5
 \end{aligned}$$

- ③
- Path 1 = 1-2-3-11-12
- Path 2 = 1-2-4-5-11-12
- Path 3 = 1-2-4-6-7-11-12
- Path 4 = 1-2-4-6-8-9-11-12
- Path 5 = 1-2-4-6-8-10-11-12

<u>Test Case ID</u>	<u>Input</u>	<u>Expected message</u>	<u>Remarks</u>
TC01	un = -50	Invalid input	Path 1
TC02	un = 95	bill = 71.25	Path 2
TC03	un = 299	bill = 274	Path 3
TC04	un = 500	bill = 712.5	Path 4
TC05	un = 501	bill = 1008	Path 5

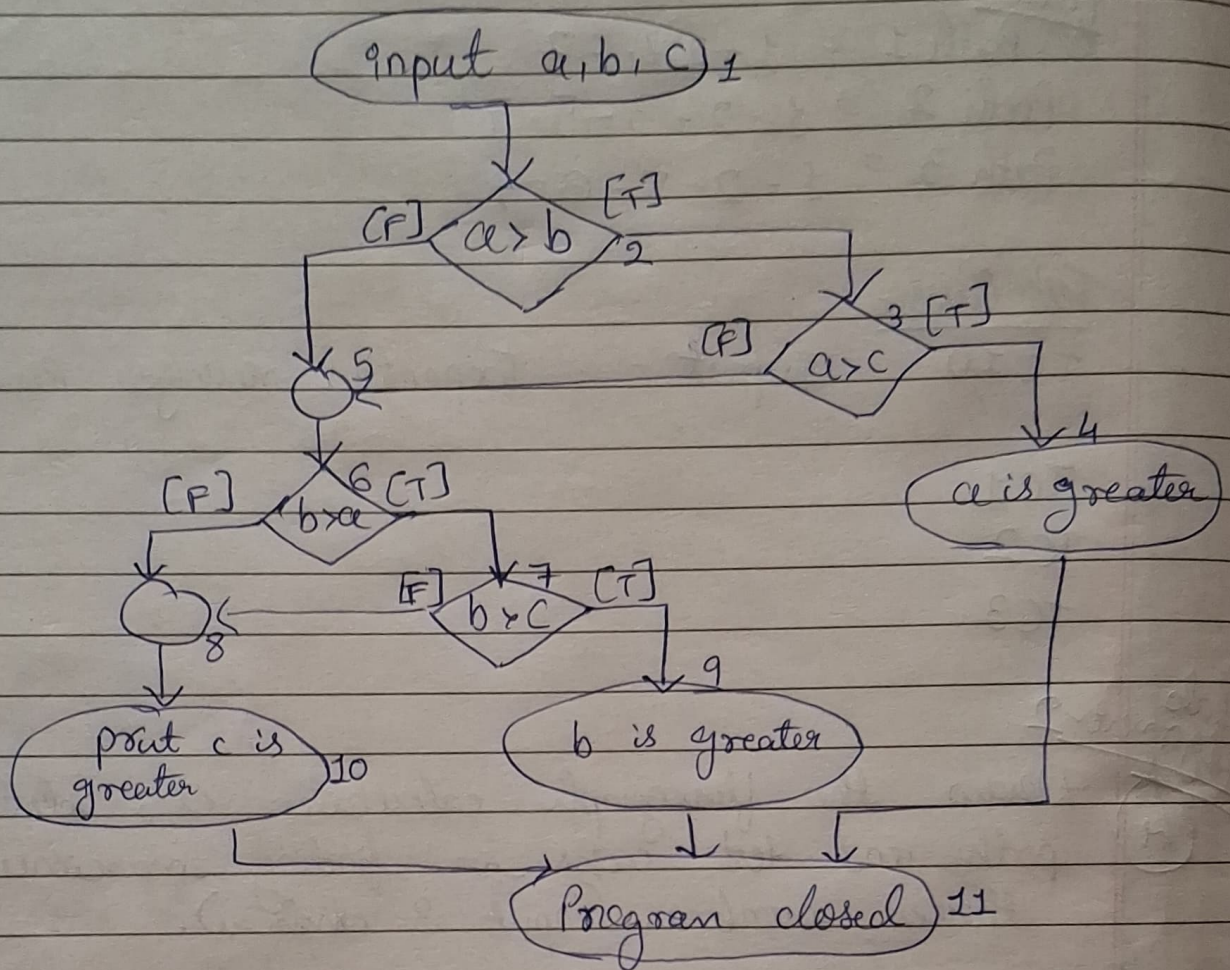
Lab Assignment-2

4) Draw the flowgraph, calculate CC, identify the path and test cases for finding maximum for three numbers. (Input 3 numbers).

Pseudo
code

```
int a, b, c;  
printf("Enter three numbers: ");  
scanf("%d %d %d", &a, &b, &c);  
if (a >= b && a >= c)  
    printf("a is greater");  
else if (b >= a && b >= c)  
    printf("b is greater");  
else  
    printf("c is greater");
```

Flow graph



CC5

$$CC = E - N + 2$$

$$= 14 - 11 + 2 = 3 + 2 = 5$$

Path 3 = 1-2-3-5-6-7-8-10-11

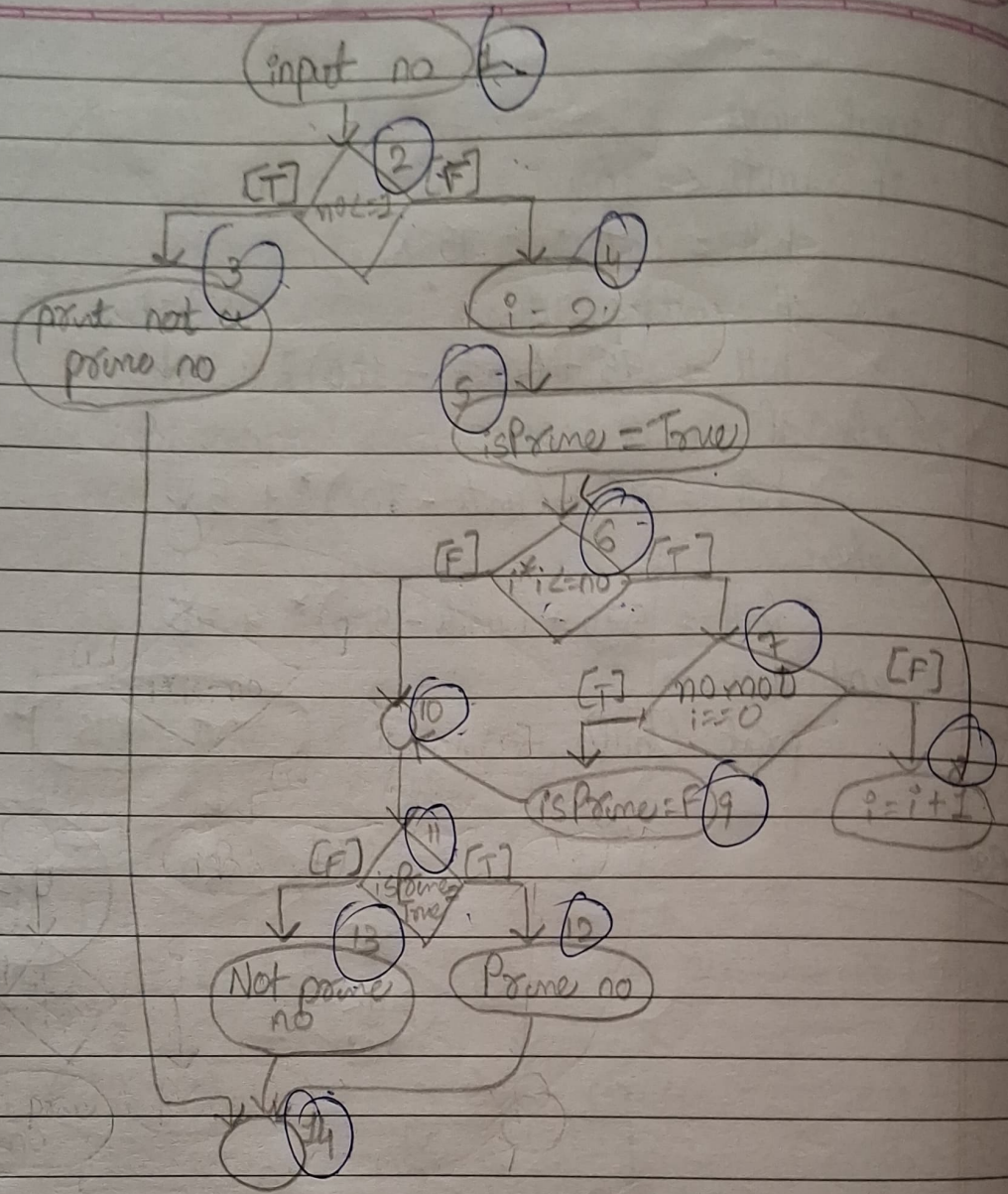
Path 4 = 1-2-5-6-7-9-11

Path 5 = 1-2-5-6-7-8-10-11

Path 6 = 1-2-5-6-8-10-11

<u>Test Case ID</u>	<u>Input</u>	<u>Expected message</u>	<u>Remarks</u>
TC01	a=5, b=6, c=8	c is greater	Path 5
TC02	a=8, b=5, c=4	a is greater	Path 1
TC03	a=8, b=4, c=10	c is greater	Path 6
TC04	a=10, b=20, c=6	b is greater	Path 4
TC05	a=5, b=30, c=50	c is greater	Path 3
TC06	a=100, b=50, c=80	a is greater	Path 1

5



$$CC = E - N + 2 = 16 - 14 + 2 = 2 + 2 = 4$$

Paths

Path 1 = 1-2-3-14

2 = 1-2-4-5-6-10-11-13-14

3 = 1-2-4-5-6-10-11-12-14

4 = 1-2-4-5-6-7-8-~~10~~ 6-10-11-13-14

5 = 1-2-4-5-6-7-8-~~10~~ 6-10-11-12-14

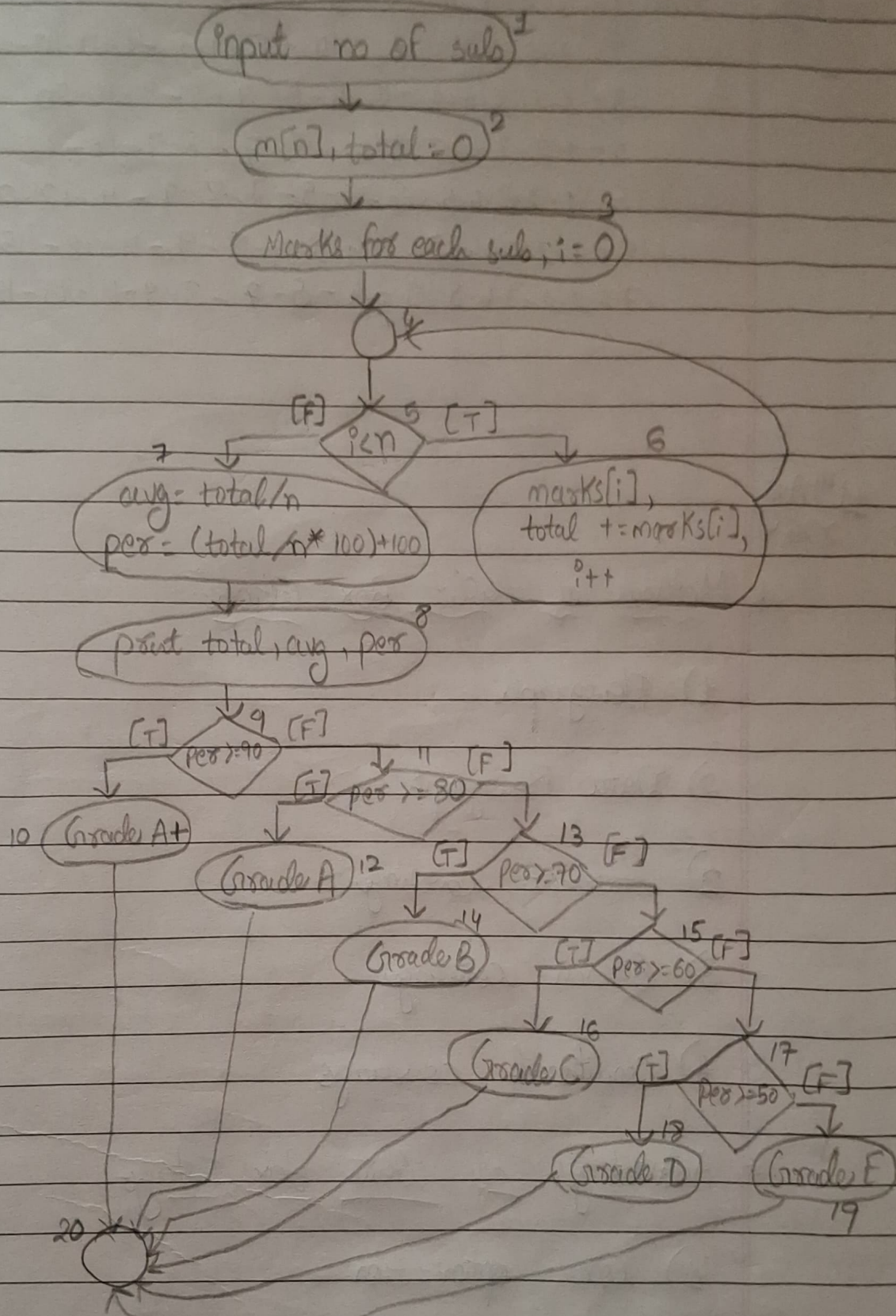
6 = 1-2-4-5-6-7-9-10-11-12-14

7 = 1-2-4-5-6-7-9-10-11-13-14

* Test Cases:

⑤	Test Case <u>ID</u>	<u>Input</u>	Expected <u>message</u>	<u>Remarks</u>
	TC01	no = 0	Not a prime no.	Invalid input Path 1 selected
	TC02	no = 5	Prime no	Valid Path 3 selected
	TC03	no = 10	Not a prime no	Valid Path 4 selected
	TC04	no = 2	Prime no.	Valid Path 3 selected
	TC05	no = -5	Not a prime no	Invalid Path 1 selected
	TC06	no = 21	Not a prime no	Valid Path 4 selected
	TC07	no = 17	Prime no.	Valid Path 5 selected

⑥



$$\begin{aligned}
 CC &= D + 1 \\
 &= 6 + 1 \\
 &= 7
 \end{aligned}$$

- ⑥ Path 1 = 1-2-3-4-5-7-8-9-10-20
Path 2 = 1-2-3-4-5-7-8-9-11-12-20
Path 3 = 1-2-3-4-5-7-8-9-11-13-14-20
Path 4 = 1-2-3-4-5-7-8-9-11-13-15-16-20
Path 5 = 1-2-3-4-5-7-8-9-11-13-15-17-18-20
Path 6 = 1-2-3-4-5-7-8-9-11-13-15-17-19-20
~ Path 7 = 1-2-3-4-5-6-4-5-7-8-9-10-20
~ Path 8 = 1-2-3-4-5-6-4-5-7-8-9-11-12-20
~ Path 9 = 1-2-3-4-5-6-4-5-7-8-9-11-13-14-20
~ Path 10 = 1-2-3-4-5-6-4-5-7-8-9-11-13-15-16-20

- ~ Path 11 = 1-2-3-4-5-6-4-5-7-8-9-11-13-15-17-18-20
- ~ Path 12 = 1-2-3-4-5-6-4-5-7-8-9-11-13-15-17-19-20

Test Case ID	Input	Expected message	Remarks
TC01	n = 3 m1 = 90, m2 = 90, m3 = 90	Total = 270 Percentage = 90% Grade A+	Path 7
TC02	n = 3 m1 = 60, m2 = 40, m3 = 35	Total = 135 Percentage = 45% Grade E	Path 12
TC03	n = 0	Grade E	Path 12
TC04	n = 3 m1 = 80, m2 = 60, m3 = 70	Total = 210 Percentage = 70% Grade B	Path 9
TC05	n = 3 m1 = 50, m2 = 60, m3 = 70	Total = 180 Percentage = 63% Grade C	Path 10
TC06	n = 3 m1 = 40, m2 = 40, m3 = 40	Total = 120 Percentage = 40% Grade D	Path 11
TC07	n = 3 m1 = 84, m2 = 85, m3 = 80	Percentage = 83% Total = 255 Grade A	Path 8
TC08	n = 2 m1 = 84, m2 = 30	Total = 114 Percentage = 57% Grade D	Path 11